

Course Title	Operating Systems	Course Code	CS2018
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Research Literature (WK8)	Credit Hrs.	3+1
Knowledge Area	Computer Science (KA02)	Grading Scheme	Relative
HEC Knowledge Area	Breadth Cores	Applicable From	Spring 2023
SDG	4 Quality Education	CEP	1
Pre-requisite(s)	CS2002, CL2002	PBL	1

Course Objective	To understand the foundations of operating system. In particular to discuss processes and threads, concurrency, scheduling, memory management, file systems, device management, and input output techniques.
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No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

A = Assignment, Q = Quiz, M = Midterm, F=Final

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Identify building blocks of an operating system.	Q1, M1	C1	1
2	Compare elements of processes and threads	A1, M1	C2	1
3	Analyze memory management techniques	Q2, M2, F	C4	2
4	Appraise characteristics of file systems	Q3, F	C4	2
5	Appraise elements of input output techniques	A2, F	C4	2

Text Books	Title	Modern Operating Systems
	Author	Andrew S. Tanenbaum
	Publisher	Pearson, 4th Edition
Reference Books	Title	Operating Systems Design and Implementation
	Author	Andrew S. Tanenbaum
	Publisher	Pearson, 3rd Edition

Week	Course Contents/Topics	Chapter*	CLO*
01	Introduction, OS concepts, system calls, OS structure.	1	1
02	OS structure (monolithic, layered, microkernels, VMs)	1	1
03	Process description and control	2	2
04	Threads, symmetric multi processing, IPC (mutual exclusion and critical section)	2	2
05	Lost update problem, busy waiting vs blocking. Peterson's algorithm	2	2
06	IPC (semaphores, mutexes) Process Scheduling	2	2
07	Memory management, paging and segmentation	3	3
08	Virtual memory, page replacement algorithms.	3	3
09	Design issues for paging systems	3	3
10	Address translation and Page fault handling	3	3
11	Files, directories and their organization	4	4
12	File system implementation and management.	4	4
13	File systems optimization	4	4
14	I/O devices, I/O hardware and I/O software	5	5
15	I/O buffering, clocks and user interface	5	5
16	RAID	5	5

*Reference book chapters are given in brackets

Assessment Tools	Weightage
Quizzes (3), Assignments (3)	20.0%
Midterm (I+II)	30.0%
Final Exam	50.0%